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Conference Paper · November 2017

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Development of a game for History Course of Secondary School Students

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Abstract—Computer Games have changed the world as they influence our culture and lives. Games with educational purposes are being incorporated in almost every field of life. These games help to learn the concepts in interesting and interactive ways. A lot of research is being conducted on Digital Game Based Learning, with purpose to study existing strategies and propose different models to improvised Learning. This paper presents the development of a game prototype for history based courses for the secondary school kids. The purpose of this prototype is to improvise learning of students in these courses.

Keywords—observation, preliminary interview, DGBL, cognitive walkthrough

I. INTRODUCTION

Digital games are being designed and developed by different researchers in different fields of life. In this era of technology, almost all the activities are being digitized, but still a lot of work and assessment are required for educational games[1][2]. According to the report by Federation of American Scientists in 2006, games offer powerful affordances for education[3]. It is suggested that collaboration with educational community can help build much powerful tools for the students[4].

Several studies have pointed out the history-based video games as an effective means of engaging students beyond traditional methods of history instruction[5][6], especially when applied in combination with skilled teaching, sound instructional design, and careful implementation[7][8]. Games along with entertainment are interactive as they give challenges to the user to achieve the goal[9]. This paper also takes a step toward implementation of a game for the education purpose. The scope for this research is the secondary school students. Game is based on the History based content, which is generally difficult for students to pay attention in class and also to memorize the contents.

II. METHODS

In order to develop the game, it was important to understand the educational context fully[10]. For the purpose of this research, three schools from the city of Hyderabad, Pakistan were considered. As the focus of the research was on Secondary school students, so class 8th was considered for this research. Fig. 1 shows the methods used to develop the game prototype. Initially classrooms were observed to identify the current teaching strategies. After observations, preliminary interview were conducted from teachers of history course to identify the needs that what should be added in the technological intervention. With the suggestions from teachers, a prototype was developed which was shared with teachers once completed and cognitive walkthrough was conducted to identify what more could be added to make it a better version. These all methods are discussed in detailed in next sections.

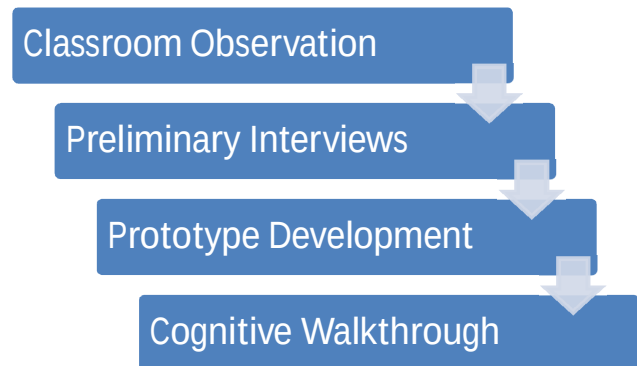


Fig. 1. Prototype Development Steps

III. CLASSROOM OBSERVATION

An observation is an act of viewing something and also at the same time receiving or acquiring knowledge from the observed environment. It is useful as it provides researchers with ways to check for nonverbal expression of feelings, determine who interacts with whom, grasp how participants communicate with each other, and check for how much time is spent on various activities[11]. Classroom observation plays

important role in identifying the teaching style, student's behaviour and learning environment[12].

For the purpose of the research, three schools were randomly selected from the city of Hyderabad, Sindh, Pakistan. This study focused on the teaching strategy being used by the teachers in the classroom, student's involvement and their learning. In this regard, meetings were conducted with heads of the schools and a formal request for conducting research was made to them. Once the request was approved, the first step towards research was conducted i.e. classroom observation. In each school, history course classroom of class 8th was observed. The normal duration of a class is 50 minutes. These were direct observations by the researcher. Table 1 shows the summary of the key factor identified during observation. The main idea behind observation was to identify the teaching strategy, use of resources and students behaviour during lectures. It was observed that almost all the classroom teaching was one-sided i.e. lecture based which had least involvement from students. Students were just listener and at the end of the lecture only one teacher conducted a paper based activity in which he asked students to write causes and impacts of wars. These observations helped to develop a deeper understanding of classroom culture in schools in this area.

Table 1. Findings from classroom observation

School	# of students in 8 th Grade	Resources/material used	Teaching strategy	Activity
A	37	Board & Marker	Lecture based	Paper based
B	18	Board & Marker	Lecture based	None
C	30	Board & Marker	Lecture based	None

IV. PRELIMINARY INTERVIEWS

After class observation, preliminary interviews were conducted from eight history course teachers from three schools. These teachers were selected because they were teaching history based courses in their schools. These teachers were identified with the help of the school management. In designing the interview, the interest was to solicit information from a teacher's perception for introducing technological perception for conducting the history course class. The focus was on teaching processes and strategies to conduct class and what are the ways they assess the learning of students and mainly to identify what are their requirement/suggestions to create a different teaching environment. A structured set of open ended-interview questions were devised with this in mind. An open-interview schedule was specifically used as it is more flexible and sensitive to issues that the respondent feels are important[13][14]. However, the time spent mainly depended on time constraints and the amount of detail given in the responses. The interview contained two major sections. The first type of questions gave information about the existing

classroom teaching pattern, strategies and activities. On the other hand, second type of questions asked about the difficulties they face in order to gain students engagement in class and their opinions and views to improvise the learning. The structured and open-ended framework ensured consistency and allowed the teachers to talk freely about their own experiences. It also gave the interviewer an opportunity to probe further if necessary[15]. To avoid a situation where the researcher would end up imposing views of the cultural situation, the researcher talked less so that the respondents talked more. All the interviews were audio taped with consent from the subjects and later transcribed. Tape recording was done to ensure completeness of the verbal interactions. At a later stage this also provides material for reliability checks. All questions were in English, and all respondents were expected to answer back in English. Table 2 highlights the main findings from the interview in order to identify the requirements for the development of a game i.e. difficulties they face in maintaining student's attention and suggestions for the improvement of learning. Responses from the teachers clearly show that the main problem in history classes is to manage their engagement level and motivate them to study. Teachers say that students feel bore as this is tough and dry course. Along with that, teachers suggested that there should be some different approach for classroom teaching beside traditional lecture based learning.

V. PROTOTYPE DEVELOPMENT

This section describes the system prototyping development of a game which includes software development requirements and user interface of the game.

A. Software Development Requirements

The game prototype is developed considering the views from teachers during the preliminary interviews from teachers. The prototype has been developed using two platforms i.e. Java and XML. List of applications or tools required towards development of game for each is as follows:

1) Platform 1 (Java):

- Eclipse IDE: system Builder
- Jasper: for generating reports
- Text to speech : Third party library marytts

2) Platform 2 (XML):

- Notepad: system Builder
- XML: to store game contents
- Jasper: for generating reports

B. User Interface of the game

In this section, interface of the game is presented. For the purpose of the research three topics from the 8th grade Social studies book were selected i.e. "Allama Iqbal", "Muslim Invasion in the sub-continent" and "Import and Export of Pakistan".

Table 2. Responses of two questions from preliminary Interview

		Interview Questions	
School	Teacher	Question # 5: Do you face any difficulties you face while teaching history based course	Question # 6: Suggestions to improve learning for history based course
A	1	Very difficult to make students understand everything	I feel if some technological advancement be given, it could be of help. I did not find any game online
	2	Students are less interested in lecture based session	I think if there will be some game or activity then students like to attend this class more than any other class
	3	It is very difficult because we just collect knowledge and there is nothing entertaining so students feel bore	Modernization is to give some technological tactics to students so they can take more interest
B	1	Mostly students are busy with their stuff and difficult to maintain their attention	You can add geographical activity or computer based activity
	2	very dry subject, there are too much dates, so many names which makes it very confusing for students	if there could be added some game or activity , students might get benefit
C	1	They don't understand main points though we give them lectures	There should be activities for students to make maps, models, globes etc
	2	Sometimes theory is too landy, so I cannot teach them well	out door activites could be helpful for the students
	3	too much difficulty, its very tough and dry subject	it should be though multimedia

1) *Log-in Interface:* Fig. 2 shows the log-in screen of the game prototype. This Screen requires students to enter their name, roll# and select the school name from the list of institutes. Once student enters the details, he/she can click start button and go for the lectures. Also this screen shows a print icon, this icon is used here to generate report for student's performance. Also there is an exit button, which student at screen, can use to exit the environment.

2) *Topic Selection:* As the users selects start button on log-in screen, the screen shown in fig. 3 appears. This screen shows as well as announces that any topic could be selected to start the lecture. As the prototype uses text to speech conversions, audio instructions are also given along with the screen.

3) *Lecture Interface:* In order to start the lecture, user needs to select one of the topics displayed. When user selects a topic, the next screen appears starting to display the contents of the topic as shown in fig. 4a and fig. 4b. Topic relevant images along with the textual content are displayed. For more

understanding of student, audio is also played which is more elaborative then the text written. Audio sounds also contain examples and meanings of the difficult words.

There are two navigation buttons showing right and left arrow. There are used to navigate through the front and back screens, in cased if a student wants to review something again, then it can select the back navigation button. Also there is an exit button which student can select anytime to exit the game. Below exit button, there is a black colour icon, which shows to change student at any time. If a student wants to change the topic during a lecture, it can be done by clicking the white board icon next to the change student icon. Using this icon, student will get the topic selection screen, from where new topic for the lecture can be selected. One last button on the screen, in which a student is sitting on the table, represents the practice test icon. Students at any stage can start practice questions, and also during the practice test they can return back to slides in case they want to review.

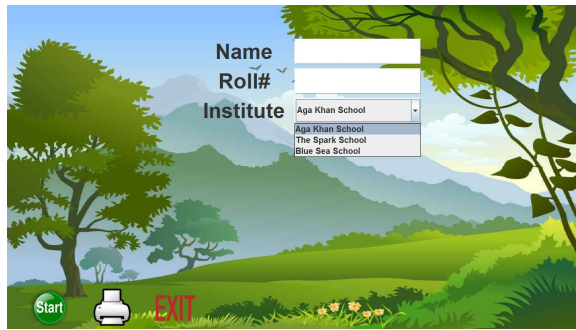


Fig. 2. Log-in Screen Interface

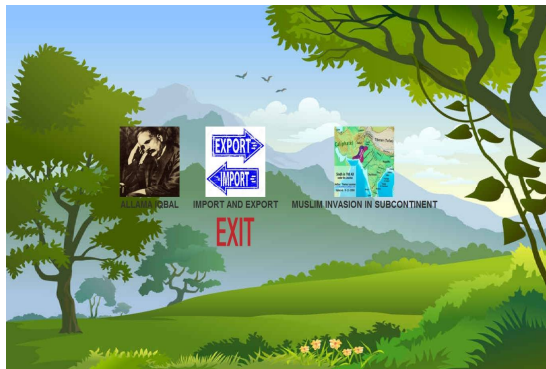


Fig. 3. Topics Selection Screen



Fig. 4a. Contents Screen

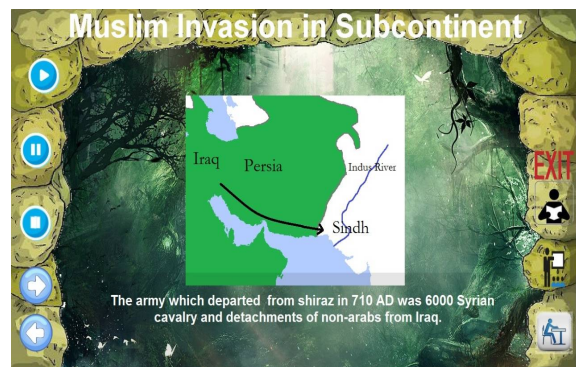


Fig. 4b. Contents Screen

4) **Practice Test:** Fig. 5 represents the practice test session. Practice test questions have been created for each topic added in the prototype. Different styles of questions have been selected like a) selecting best answer, b) selecting all the appropriate answers and c) select the appropriate picture by listening the sound clip. Emojis have been used in order to create attraction for the students so that environment can be made more interactive. There is also an emoji showing that it is thinking that what could be the answer. Also fig. 5 shows an emoji that is showing excitement when the correct answer is given. Similarly if a wrong answer is given, there is a sad emoji. Students can attempt practice test any number of times they want. Also, every time answer options are shuffled, so that more effective learning can be achieved. Frequency of practice test sessions performed by each student is recorded. All practice tests sessions.

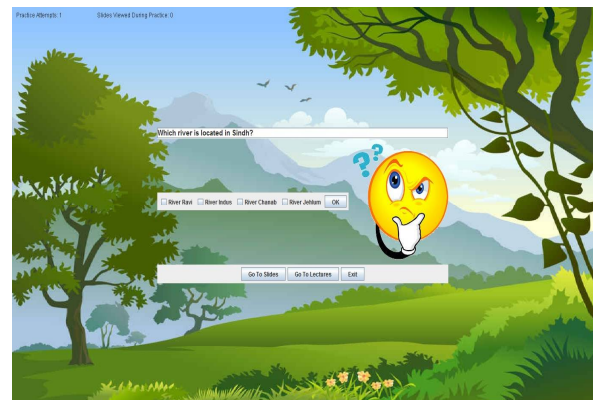


Fig. 5. Practice test screen

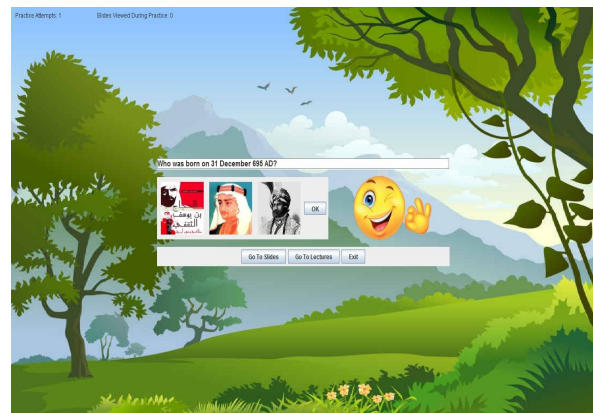


Fig. 6. Practice Test Screen with Correct Answer

5) **Reports:** The prototype maintains the statistics of student's usage of the game. There reports are generated using JasperReports API in Java language, which are exported to MS Excel. The idea is that teacher can any time view the logs and statistics of the student's interaction with the game. Fig. 7

represents the format of the report generated. They are generated school-wise.

The Spark School				
Name		Roll Number		
Chaffar		1		
Lecture #	Date	# Hits in Slides	# No. Of Practice	Time
1	24/05/2016	2	2	20:01:36
Imambux		2		

Fig. 7. Statistics of students performance

VI. SYSTEM PROTOTYPING EVALUATION: COGNITIVE WALKTHROUGH

Cognitive walkthrough is a task-specific method of imagining people's thoughts and action when they use interface for the first time. This method was developed by Wharton et al. in 1994. It is a cognitive task where focus is on the usability of an interface along with the ease of use. The walkthrough may be performed without real users and It can be done on design, prototype or mockups[16]. Walkthroughs identify problems that new users will have when they first use an interface. The walkthrough will often identify many more problems than you would find with a single, unique user in a single test session.

During a cognitive walkthrough, evaluator follows various scenarios. The evaluator takes the part of the primary stakeholder and tries to confirm the completeness of the design and robustness of the system

A. Participants

For the cognitive walkthrough, similar participants as that of preliminary interview i.e. eight teachers from three identified schools were selected. These were selected because they have been the part of the evaluation study and they know the idea better

B. Purpose of Study

The purpose of this walkthrough was to make teachers familiar with the new interface that is given to their students. The idea was to determine whether the proposed prototype is efficient and effective for the classroom learning and also that will be able to engage and motivate students? Efficiency refers to the ability to allow participants to carry out their activities in a smooth manner and that will help us to engage students with the system. Effectiveness here means that using this prototype there is improvement in learning. If their learning is

improved, they will be motivated to use such systems, as this would be interactive and at the same time improving their class performance. It was expected that cognitive walkthrough by teachers would help identify the problems in the interface/design. Once the teachers identify the problems, they are incorporated before performing the experiments with the real users i.e. the students of the course.

C. Evaluation Procedure and Design

This section describes the briefing procedure, task descriptions and list of tasks performed by the participants.

1) *Briefing on the Game Prototype:* At first, Participants were greeted and introduced to the game using the developed prototype. The overall duration of the cognitive walkthrough session took about 2 hours in each school. This walkthrough was conducted in each school separately with their teachers. The initial 15 minutes were spent in giving a brief explanation of the prototype and the purpose of this experiment. A glimpse of briefing text is shown in fig. 8. Participants spent the rest of their time in experiencing the game by performing given list of tasks and actions. Next section describes the task description along with the action sequence to complete the task.

I would like to emphasize that this is not a test, instead this is an evaluation study to make things work in improvised form. As you all have been part of the preliminary interview, so you all know in detail the purpose of my research. The purpose is to give a new approach to classroom learning besides the traditional teaching style. Today the study we are going to conducted is based on what you all suggested in the interview. A game prototype has been developed. Before we move on, I should tell you what a prototype is? A *prototype* is an early sample, model, or release of a product built to test a concept or process.

The study we going to perform is cognitive walkthrough. It is a task-specific method of imagining people's thoughts and action when they use interface for the first time. Similarly, as you will use the interface today for the first time. You have been provided a paper, containing

Fig. 8. Debriefing Participant Script

2) *Task Description and List of Actions:* The tasks are developed based on the needs of activities to be carried out in order to perform experimentation. Fig. 9 represents three main tasks which are selected for this evaluation study. After each task is performed, the participant were asked following questions:

- Was the user able to achieve desired action?
- Did the action show its desired effect?
- How was the flow of these actions?
- Do you face any difficulty?
- Do you have any recommendations for the improvement?
- Any other comments?

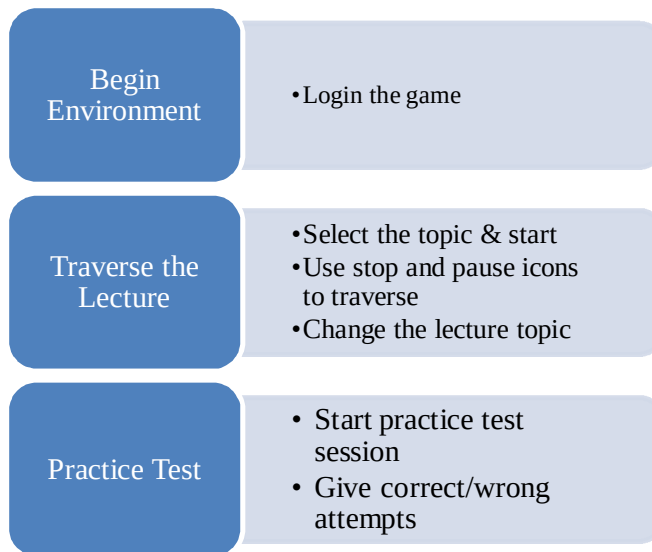


Fig. 9. Main tasks in Cognitive Walkthrough

3) *Think Aloud Technique*: As the purpose of this evaluation is to improvise the prototype before giving it to real users, it is important to know what participants think of the system. In order to get more appropriate data, think-aloud technique is being used in this study. As the name suggests, it allows evaluators to say out loud what they are thinking about any writing, design, system etc. In this data gathering technique, participants are asked to say whatever comes into their mind as they complete the task. This might include what they are looking at, thinking, doing, and feeling. This technique gives observers insight into the participant's cognitive processes (rather than only their final product), to make thought processes as explicit as possible during task performance.

For this evaluation study, think-aloud technique is selected so that evaluation at each step can express their views, observations and findings. It also gives participants to discuss aloud their reasons, decisions or suggestions while performing tasks. The data collected from participants for this study is qualitative. This data is collected in three phases as the study was divided in three main tasks. The summarized data from the study is shown in next section.

D. Data Results and Findings

In this section, observations and suggestions for the improvement of game prototype are presented in three phases i.e. main tasks for the study.

1) *Begin Environment Task*: In this task, participants initiated the game environment and performed the log-in task. The participant conveyed that the environment start screen seems attractive. It increases interest to play it and quickly go through it. They evaluators felt that as students might make spelling mistakes, the name of the school should appear as a list.

2) *Traverse the Lecture*: In the traverse the lecture task, as participant selected the topic the contents for that topic along

with the sound, subtitles and images appeared on the screen. The participants stated that sound clip is very helpful to learn and memorize the contents effectively. They also experience the pause and stop feature to play the contents repeatedly. They expressed that this feature helped them to understand the contents and more.

Evaluators felt the need that there should be next and previous traverse option as well in order to move to previous contents. In current situation, if a user needs to view previous slide, it needs to restart the lecture. Another observation that was highlighted from almost all the participants was that the text appearing on screen with too many animations could not help them to be keeping their focus on the contents. Character wise animations were added which seemed annoying to the participants. Also observations suggested that animation were making the content appear slower, so it is hindrance to effective readability.

3) *Practice Test*: In this task, participants selected the practice test icon from the lecture view. Once the practice test was started, they liked the variety of questions added. As variety of question gain their attention and interest more. The need was felt that if a user enters correct or incorrect information, there should be some animation or icon that may represent that question is correct or incorrect.

E. Discussion

The cognitive walkthrough has been proved very effective in order to take out feedback from the subject teachers for the improvement of the game. Study shows that overall evaluation of game is satisfactory and accepted by teachers. Participants suggested few changes that need to incorporate to make game more effective. Table 3 summarizes the suggestions by the participants of the study. All the suggestion were incorporated as per the requirements by the teachers.

Table 3. Cognitive Walkthrough - Suggestions

Main Tasks	Suggestions for Improvement
Begin Environment	1. Names of the schools be added as a list
Traverse the Lecture	1. Animation is slow 2. There should be character wise animation, instead text should appear as a line or paragraph 3. Navigation for front and back screens be added
Practice Tests	1. Icons may be used to represent the answers correct or incorrect state

VII. CONCLUSION

This paper discusses the game prototype development for the classroom learning for secondary school kids. In order to develop prototype, classroom observations and preliminary interview were conducted, which have presented in great

detail. In order to measure the impact of game on learning of students, it is important to test the game with the students. Researcher intends to perform different experiments with students in real time classroom environment.

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